

Thomas Traversié

PhD student in Computer Science

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Research interests

- ⊢ Proof theory
- ⊢ Type theory
- ⊢ Logical frameworks
- ⊢ Proof translations

Education

- Since 2023 **PhD student in Computer Science**, *Université Paris-Saclay*, Gif-sur-Yvette
MICS and LMF laboratories
Supervised by Marc Aiguier, Gilles Dowek and Olivier Hermant
Topic: Translation of proofs between higher-order logics and between theories with rewriting
- 2020–2023 **Engineering degree**, *CentraleSupélec*, Gif-sur-Yvette
Major in Computer Science – Software Sciences
Research Track
- Febr.–July 2022 **Academic semester**, *École Polytechnique Fédérale de Lausanne*, Switzerland
International exchange in the Master's program in Computer Science
- 2018–2020 **Classe préparatoire MPSI/MP***, *Lycée Montaigne*, Bordeaux
Intensive preparatory training in Mathematics and Physics, option in Computer Science
- 2018 **Scientific Baccalauréat**, *Lycée Palissy*, Agen

Experience

- 2023–2026 **Teaching activities**, *CentraleSupélec*
I have taught 196 hours at the CentraleSupélec engineering school at the bachelor and master levels. I gave lectures, exercise sessions, lab sessions and project sessions. I participated in both basic computer science courses (algorithmics, programming, information systems) and specialized courses (logic, automata, formal languages, theoretical computer science).
- April–Sept. 2023 **Research internship**, *Inria Saclay and LMF*
Deducteam team, supervised by Valentin Blot and Gilles Dowek, collaboration with Théo Winterhalter
Topic: Replacing rewrite rules by axioms in the $\lambda\Pi$ -calculus modulo rewriting
Master thesis: Rewrite Rules in the $\lambda\Pi$ -Calculus Modulo Theory
- 2020–2023 **Research track**, *Inria Saclay and LMF*
Deducteam team, supervised by Valentin Blot and Gilles Dowek
The Research Track of CentraleSupélec allows students interested by research to carry out a research project in a laboratory, half a day per week
Topic: Implementing set theory with pointed graphs in the Dedukti proof system

List of publications

Peer-reviewed international conferences

- FSCD'25 **Monad Translations for Higher-Order Logic**, *Thomas Traversié*, 10th International Conference on Formal Structures for Computation and Deduction
[\[PDF\]](#)
- FoSSaCS'24 **From Rewrite Rules to Axioms in the $\lambda\Pi$ -Calculus Modulo Theory**, *Valentin Blot, Gilles Dowek, Thomas Traversié and Théo Winterhalter*, International Conference on Foundations of Software Science and Computation Structures
[\[PDF\]](#)

Peer-reviewed international workshops

- LFMTP'24 **Kuroda's Translation for the $\lambda\Pi$ -Calculus Modulo Theory and Dedukti**, *Thomas Traversié*, International Workshop on Logical Frameworks and Meta-Languages: Theory and Practice
[\[PDF\]](#)
- LFMTP'24 **Proofs for Free in the $\lambda\Pi$ -Calculus Modulo Theory**, *Thomas Traversié*, International Workshop on Logical Frameworks and Meta-Languages: Theory and Practice
[\[PDF\]](#)
- LFMTP'22 **An Implementation of Set Theory with Pointed Graphs in Dedukti**, *Valentin Blot, Gilles Dowek and Thomas Traversié*, International Workshop on Logical Frameworks and Meta-Languages: Theory and Practice
[\[PDF\]](#)

Submitted papers

- 2026 **Investigations on Higher-Order Infinitary Logic**, *Thomas Traversié, Olivier Hermant and Marc Aiguier*
- 2025 **Formalizing Representation Theorems for a Logical Framework with Rewriting**, *Thomas Traversié and Florian Rabe*
[\[PDF\]](#)
- 2024 **Kuroda's Translation for Higher-Order Logic**, *Thomas Traversié*
[\[PDF\]](#)

Software development and implementation

TranslationTemplates, allowing the translation of proofs between Dedukti theories, using three translation templates, written in OCaml

Construkti, allowing the translation of Dedukti proofs from classical logic to intuitionistic logic, written in OCaml

Library of proofs in Lambdapi, encoding set theory using pointed graphs

Talks

- January 2026 Morphisms between Dedukti Theories, ICSPA project meeting
- October 2025 Monad Translations for Higher-Order Logic, Journées GT Scalp 2025

July 2025 Monad Translations for Higher-Order Logic, FSCD 2025
 Febr. 2025 Translation Templates between Dedukti Theories, KWARC seminar
 Sept. 2024 Generic Translations between Dedukti Theories, WG1+2+4 meeting (EuroProofNet)
 July 2024 Proofs for Free in the $\lambda\Pi$ -Calculus Modulo Theory, LFMTTP 2024
 July 2024 Kuroda's Translation for the $\lambda\Pi$ -Calculus Modulo Theory and Dedukti, LFMTTP 2024
 June 2024 Kuroda's Translation for Higher-Order Logic, Logimics meeting
 June 2024 Kuroda's Translation for Higher-Order Logic, LMF summer seminar
 May 2024 Replacing Rewrite Rules by Equational Axioms in the $\lambda\Pi$ -Calculus Modulo Theory, LMF PhD seminar
 April 2024 From Rewrite Rules to Axioms in the $\lambda\Pi$ -Calculus Modulo Theory, FoSSaCS 2024
 Dec. 2023 Règles de Réécriture dans le $\lambda\Pi$ -Calcul Modulo Théorie, Master thesis defence
 August 2022 An Implementation of Set Theory with Pointed Graphs in Dedukti, LFMTTP 2022

Participation to events

Research visit

February 2025 One-week visit hosted by Florian Rabe, FAU Erlangen-Nürnberg (Germany)
 1500€ Short-Term Scientific Mission grant from EuroProofNet

Conferences and workshops

July 2025 FSCD 2025, Birmingham (United Kingdom)
 July 2024 LFMTTP 2024 et IWC 2024, Tallinn (Estonia)
 April 2024 ETAPS 2024, including ETAPS Mentoring workshop, Luxembourg (Luxembourg)
 August 2022 FLoC 2022, Haifa (Israel)

EuroProofNet working groups

I am involved in the EuroProofNet research network, that focuses on the interoperability and usability of proof systems:

Febr. 2025 WG1 meeting, Nogent-sur-Seine (France)
 Sept. 2024 WG1+2+4 meeting, Fontainebleau (France)
 January 2024 Meeting on Dedukti and proof systems interoperability, Gruissan (France)
 Sept. 2023 WG1+4 meeting, Fontainebleau (France)
 January 2023 Dedukti tools developers meeting 2, Fréjus (France)

Summer schools

Sept. 2025 International School on Logical Frameworks and Proof Systems Interoperability (LFPSI), Orsay (France)
 August 2024 International School on Rewriting 2024 (ISR), Obergurgl (Austria)

Administrative and collective responsibilities

Since 2024 **Representative of PhD students**, *Interfaces Doctoral School*, Université Paris-Saclay

Sept. 2025 **Organizational support**, *EuroProofNet Symposium*, Orsay

Skills

└ **Languages:** French (native), English (advanced), Spanish (intermediary)

└ **Programming languages:** OCaml, Python, C/C++, Java, Scala, $\text{\LaTeX}$, Git, SQL, HTML

└ **Proof tools:** Rocq, Isabelle, Méthode B, Frama-C